

CSCI-GA.1170-003 Fundamental Algorithms

September 15, 2026

Problem Set 2

Lecturer: Julia Kempe

Due: 11:59pm on Monday, September 21

Problem 2-1 (Different Methods for Recurrences)**14 points**

Consider the recurrence

$$T(n) = 16T\left(\frac{n}{2}\right) + 5n^3$$

with initial condition $T(1) = 1$. You may assume that n is a power of 2.

- (a) **(2 points)** Solve it asymptotically using the “master theorem”.
- (b) **(4 points)** Solve it by the “guess-then-verify method”. Namely, guess a function $g(n)$ — presumably solving part (a) will give you a good guess — and argue by induction that for all values of n we have $T(n) \leq g(n)$. What is the “smallest” $g(n)$ for which your inductive proof works?
Hint: If your induction does not go through with a single leading degree term, then you may be able to use other degree terms?
- (c) **(4 points)** Solve it by the “recursive tree method”. Namely, draw the full recursive tree for this recurrence, and sum up all the values to get the final time estimate. Again, try to be as precise as you can (i.e., asymptotic answer is OK, but would be nice if you preserve a “leading constant” as well).
- (d) **(4 points)** Solve it precisely using the “domain-range substitution” technique. Namely, make several changes of variables until you get a basic recurrence of the form $S(k) = S(k-1) + f(k)$ for some f , and then compute the answer from there. Make sure you carefully maintain the correct initial condition, and check that your answer agrees with part (c).

Problem 2-2 (Fun with Recurrences)**17(+4) points**

Solve the following recurrences using any method you choose. Parts (a) and (d) ask for exact solutions outright and $\Theta(\cdot)$ bound. For parts (b) and (c) it suffices to solve the recurrence up to $\Theta(\cdot)$; two points of extra credit per part are awarded for an exact solution.

- (a) **(4 points)** Solve the following recurrence *exactly*:

$$T(n) = \begin{cases} 2 & n = 1 \\ 2^{(n/2)+1} \cdot T\left(\frac{n}{2}\right) + 3 \cdot 2^{(n+1)} & n > 1 \end{cases}$$

You may assume that n is a power of 2.*Hint:* divide both sides by something to rewrite the equation slightly,

(b) **(5 points, +2 extra credit)**

$$T(n) = \sum_{i=1}^{\log_2 n} 2^i \cdot T\left(\frac{n}{2^i}\right) + n^2, \quad n > 1, \quad T(1) = 1.$$

Hint: assume that n is a power of 2.

(c) **(5 points, +2 extra credit)**

$$T(n) = \begin{cases} 1 & n = 7 \\ 2T\left(\frac{n}{2} + 3\right) + n & n > 7 \end{cases}$$

Hint: try domain-range substitution. Assume n can be written as $2^m + 6$

(d) **(3 points)** Solve the following recurrence *exactly*:

$$T(n) = \begin{cases} 2 & n = 1 \\ 4\left(1 + \frac{1}{n}\right)^2 \cdot T(n-1) & n > 1 \end{cases}$$

Hint: consider telescoping cancellation.

Problem 2-3 (Functionality vs. Running Time)

14 points

In this question we explore the difference between the functionality of a function and the running time of the same. Recall that functionality refers to the purpose of the procedure, i.e., what is it designed to produce as an output. Additionally, we will be looking at a randomized algorithm where we will be dealing with expectations.

Expectations. Let X be a random variable that takes values $x_1, \dots, x_k$ with probabilities $p_1, \dots, p_k$. Then,

$$\mathbb{E}[X] = \sum_{i=1}^k x_i \cdot p_i = x_1 \cdot p_1 + \dots + x_k \cdot p_k.$$

Additionally, we have the linearity of expectation, i.e., for any real a, b, c and any random variables X, Y ,

$$\mathbb{E}(aX + bY + c) = a\mathbb{E}(X) + b\mathbb{E}(Y) + c.$$

Now, we consider the following recursive procedure.

BLA(n):

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1 if  $n = 1$  then return 1
2 else
3   Assign  $x$ : 0 with probability  $\frac{1}{4}$ , 1 with probability  $\frac{1}{2}$ , or 2 with probability  $\frac{1}{4}$ 
4   if  $x = 0$  then return BLA( $n$ )
5   else if  $x = 1$  then return  $10 \cdot \text{BLA}(n/3) + 3$ 
6   else if  $x = 2$  then return  $4 \cdot \text{BLA}(n/3) + \text{BLA}(n/3) + \text{BLA}(n/3) + \text{BLA}(n/3) + 3$ 
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- (a) **(4 points)** Present a recurrence equation for the expected running time of $\text{BLA}(n)$. You may assume that mathematical operations, if statements, and the value assignment of x can be accomplished in constant time. Further, assume that n is a power of 3.

Use Master Theorem to solve this recurrence relation to obtain an asymptotically tight value. Do not forget to explicitly state the values of a , b , $f(n)$ and identify the correct case of the Master Theorem, with suitable justification for full credit.

- (b) **(5 points)** Let the above function be invoked by the line $y = \text{BLA}(n)$. Now, present a recurrence equation to represent the expected value that is assigned to y by the invocation call. You may assume that mathematical operations, if statements, and the value assignment of x can be accomplished in constant time. Further, assume that n is a power of 3.

Use Master Theorem to solve this recurrence relation to obtain an asymptotically tight value. Do not forget to explicitly state the values of a , b , $f(n)$ and identify the correct case of the Master Theorem, with suitable justification for full credit.

- (c) **(5 points)** Now assume that you rewrite Line 6 of the code as

6 **else if** $x = 2$ **then return** $7 \cdot \text{BLA}(n/3) + 3$

How will the two recurrence relations change? Present the modified recurrence relations. You may assume that mathematical operations, if statements, and the value assignment of x can be accomplished in constant time. Further, assume that n is a power of 3.

Use Master Theorem to solve the modified recurrence relation(s) to obtain an asymptotically tight value. Do not forget to explicitly state the values of a , b , $f(n)$ and identify the correct case of the Master Theorem, with suitable justification for full credit.